Frontend Code Challenge - Onboarding Team

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📋 The Challenge: Multi-Step Insurance Quote Calculator

Build a **multi-step web application** that walks a user through generating an insurance quote. The flow mirrors the complexity of our onboarding flows: it involves input collection, conditional branching, calculated results, and a mock API submission.

Functional Requirements

Step 1: Personal Information

- Collect: name, email, age, and zip code.
- All fields are required and must be validated before proceeding.

Step 2: Coverage Selection

- User selects a coverage type: **Basic**, **Standard**, or **Premium**.
- Additional questions appear conditionally, if `age > 65`, show supplemental health questions:
    - **Pre-existing conditions** — "Do you have any pre-existing medical conditions?" (Yes / No). If Yes, show a multiselect: Diabetes, Heart Disease, Hypertension, Cancer (history), Other. Tests nested conditional logic.

- **Prescription medication** — "Do you currently take prescription medication on a regular basis?" (Yes / No). Tests simple boolean branching.
- **Tobacco use** — "Do you smoke or use tobacco products?" (Yes / No). Should trigger a recalculation of the estimated monthly premium. Tests state-driven premium calculation.
- **Spouse coverage** — "Do you require coverage for a spouse or domestic partner?" (Yes / No). Should be reflected in the Summary step. Tests cross-step state propagation.

- Display a calculated estimated monthly premium that updates dynamically based on selections.

Step 3: Summary & Submission

- Show a read-only review of all collected data and the calculated quote.
- A **Submit** button triggers a mock API call (simulated with a delay and success/error state).

Monthly Premium Formula

The estimated monthly premium must be calculated using the following formula:

`monthlyPremium = basePremium × ageMultiplier × conditionsMultiplier × tobaccoMultiplier × spouseMultiplier`

Base Premium by Coverage Type

Coverage	Base Premium
Basic	$50
Standard	$100
Premium	$200

Multipliers

Factor	Condition	Multiplier
Age	`age > 65`	`× 1.5`
Pre-existing conditions	Any selected	`× 1.3`

Factor	Condition	Multiplier
Tobacco use	Yes	`× 1.2`
Spouse coverage	Yes	`× 1.4`

Example: A 70-year-old selecting Standard coverage, with a pre-existing condition, who smokes and wants spouse coverage:

`$100 × 1.5 × 1.3 × 1.2 × 1.4 = $327.60/month`

Technical Requirements

Area	Requirement
Language	TypeScript
UI Framework	React
UI Component Library	Material UI (MUI)
Forms	React Hook Form + Yup for schema validation
State	Context API to share data across steps
Navigation	Multi-step with a visible progress indicator
Conditional Logic	Age-based conditional questions
Responsive Design	Must work across mobile and desktop viewports
Error Handling	Handle validation errors, failed submissions, and empty states gracefully
Testing	Unit tests for at least 2 components

Bonus Points (Optional)

- Route-based navigation using **React Router** (each step as its own route)
- Form data persistence (via `localStorage`) so users can resume progress
- Static code analyzer (ESLint + a custom ruleset or Prettier config)
- Accessibility considerations (ARIA labels, keyboard navigation, focus management)

Submission

Submit your solution as a **public GitHub repository**. The README must include:

- Setup and run instructions
- A brief description of your development process and decisions
- A description of the architecture used and the reasoning behind your choices

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